

Early Signals of Small Business Lifecycle Transitions

Author: Manus AI **Date:** April 2026

Executive Summary

This study investigates the lifecycle transitions of small businesses using high-frequency weekly transaction data. By analyzing a dataset of 6.58 million store-week observations across 59,089 stores, we aim to determine whether early customer acquisition and transaction patterns can serve as predictive signals for future lifecycle transitions.

Our findings indicate that early transaction patterns—specifically the rate of new customer acquisition and early sales momentum—are strong leading indicators of future growth or decline. Using early window data (first 30 or 52 weeks), machine learning models successfully predicted future high growth (ROC-AUC 0.87–0.95) and transaction inactivity risk (ROC-AUC 0.93). Furthermore, event study and Granger-style analyses confirm that spikes in new customer acquisition precede sustained sales growth. Unsupervised state discovery suggests a moderate cluster structure in business lifecycles, with distinct states characterized by varying levels of sales volume and customer acquisition rates.

Importantly, this study relies solely on transaction data without industry, geography, or true closure labels. Therefore, we use cautious terminology—such as "transaction inactivity" rather than "closure"—and emphasize predictive associations rather than causal effects.

1. Data Audit and Feature Engineering

1.1 Data Overview

The dataset (`weekly.parquet`) contains 6,582,263 weekly observations spanning from January 2021 to August 2023 (142 weeks). It includes 59,089 unique stores, with an average of 111 observations per store.

Key variables include:

- **Sales Channels:** Card sales, invoice sales, and delivery sales.
- **Customer Metrics:** Total customers and new customers.
- **Temporal Patterns:** Weekend sales, before-noon sales, and after-noon sales.

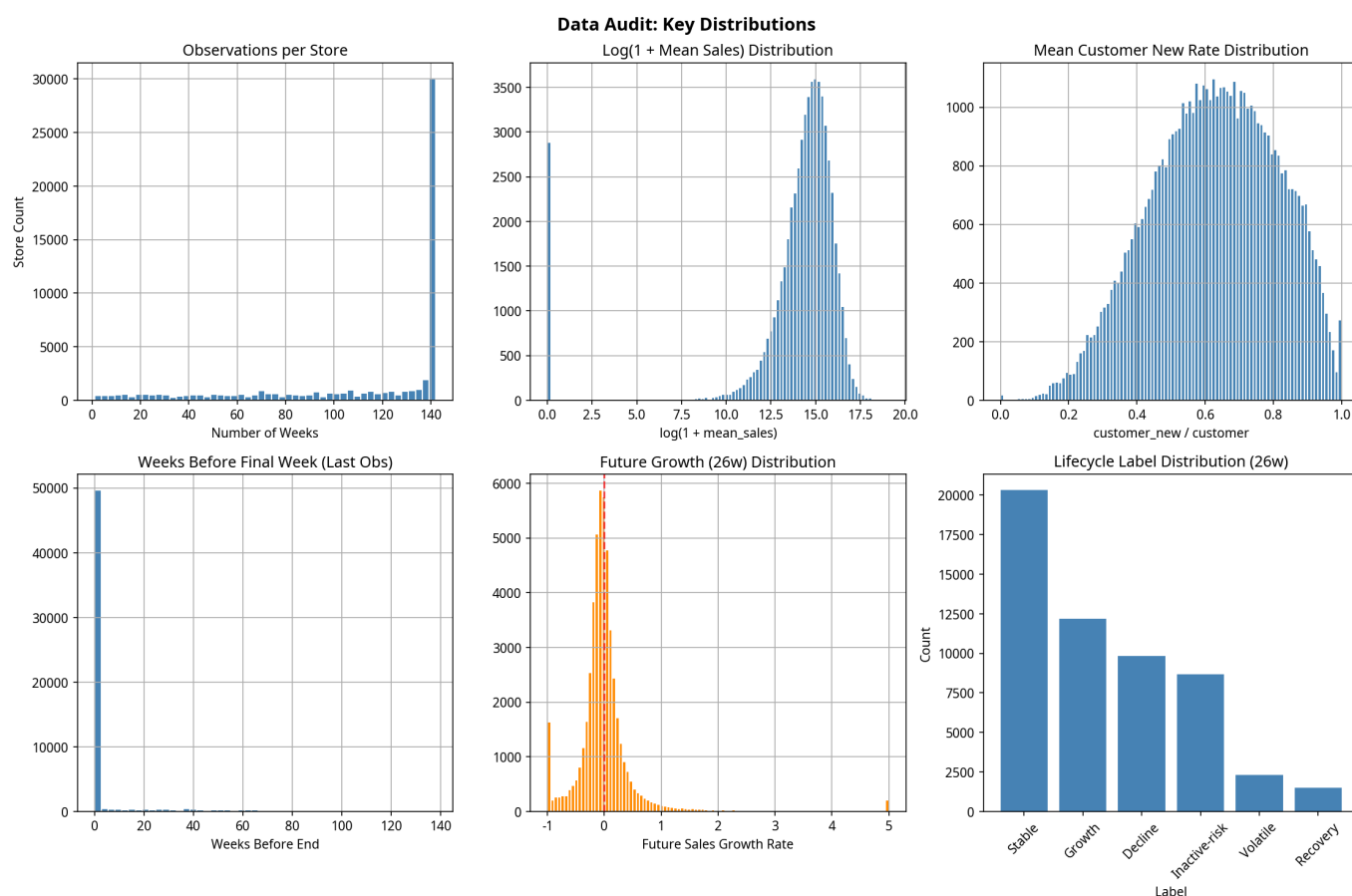
1.2 Missingness and Abnormal Values

- Card sales and customer counts have approximately 13.6% missing values, which were handled via imputation or feature engineering.
- Invoice and delivery sales have high missingness (>80%), reflecting specific business models rather than data errors.
- No negative values were found in card sales or customer counts. A negligible number of negative invoice sales (243 cases) were clipped to zero.

1.3 Feature Engineering

We constructed comprehensive store-week features to capture dynamic lifecycle patterns:

- **Rolling Metrics:** 4, 8, 13, and 26-week rolling means and standard deviations for sales and customer counts.
- **Growth and Momentum:** Sales growth over various horizons, momentum (4-week vs. 13-week mean), and acceleration.
- **Volatility and Drawdown:** Coefficient of variation (CV) over 13 weeks and drawdown from the 26-week rolling maximum.
- **Customer Acquisition:** Customer New Rate (CNR) defined as $\text{customer_new} / \text{customer}$. We also identified "CNR spikes" using store-level percentiles and rolling Z-scores.



2. Outcome Definitions and Lifecycle Labels

Due to the absence of true closure labels, we defined lifecycle transitions based on future sales performance and transaction inactivity.

2.1 Future Growth and Decline

For 13, 26, and 52-week horizons, we calculated the future sales growth rate relative to the current 13-week baseline.

- **High Growth:** Top quartile of future growth.
- **Decline:** Bottom quartile of future growth.

2.2 Transaction Inactivity

Stores with no observed transactions for 8 or 13 weeks prior to the final dataset week were flagged for "transaction inactivity risk." We explicitly avoid labeling this as "closure," as businesses may operate through unobserved channels or experience temporary seasonal pauses. Approximately 14.7% of stores exhibited inactivity of 8 weeks or more.

2.3 Composite Lifecycle Labels

Based on a 26-week horizon, stores were categorized into:

- **Growth:** 21.8%
 - **Decline:** 21.8%
 - **Stable:** 38.6%
 - **Volatile:** 3.1%
 - **Inactive-risk:** 14.7%
-

3. Leading Signal Analysis

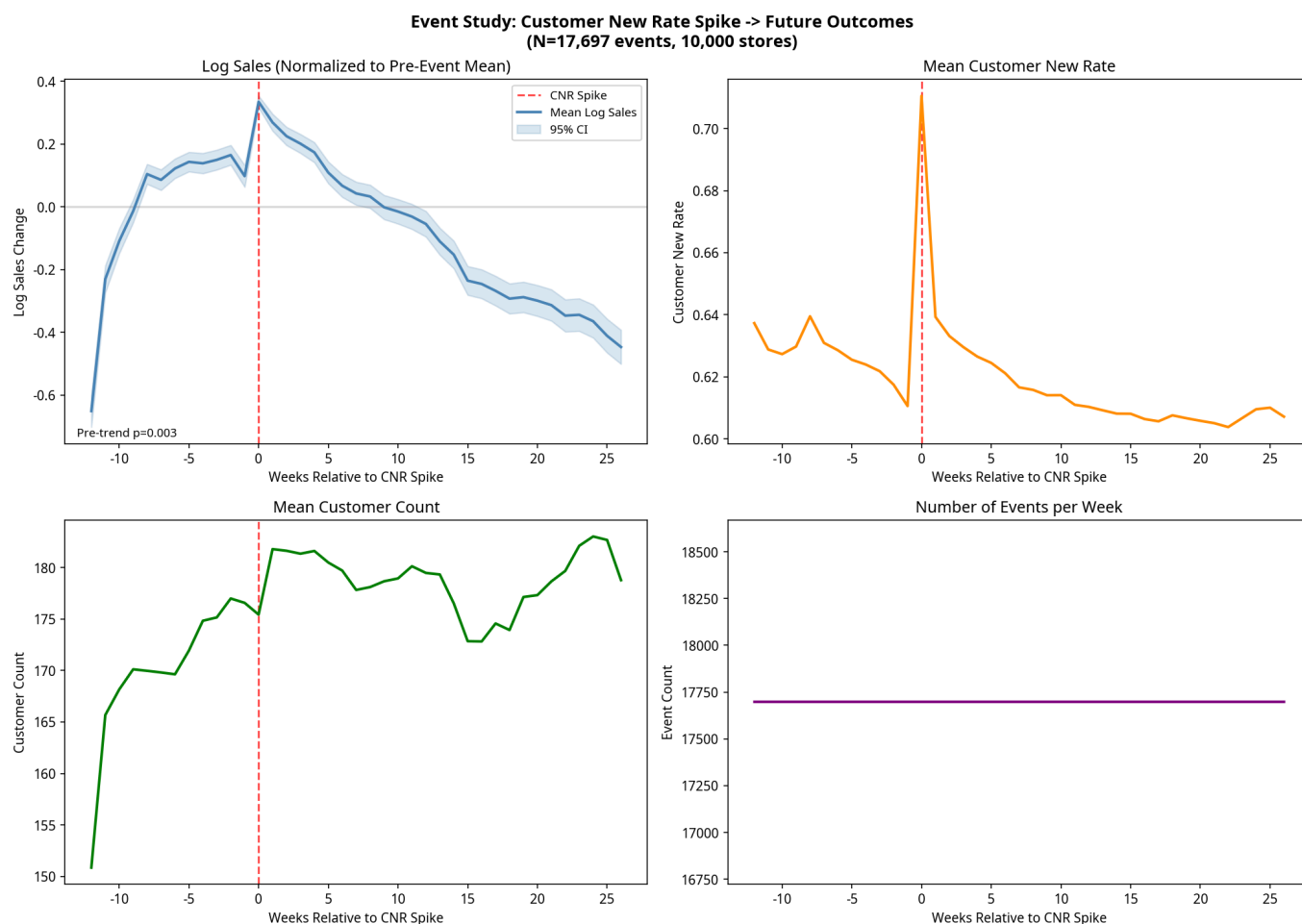
We tested whether early customer acquisition (CNR) acts as a leading indicator for future sales growth.

3.1 Event Study

We conducted an event study analyzing the 12 weeks prior to and 26 weeks following a significant spike in the Customer New Rate (CNR > 75th percentile).

- **Results:** Following a CNR spike, normalized log sales show a sustained upward trajectory.

- **Pre-trend Check:** The pre-event period showed no significant upward trend, supporting the temporal precedence of the CNR spike.



3.2 Panel Regression and Granger-Style Tests

We estimated panel regressions with store fixed effects to test the association between lagged CNR and current sales.

- **Panel Regression:** Lagged CNR (1, 4, and 8 weeks) is positively and significantly associated with log sales ($p < 0.001$). The effect diminishes by week 13.
- **Granger-Style Test:** Adding lagged CNR significantly improves the prediction of 4-week sales growth compared to a model using only autoregressive sales lags (F-test $p < 0.001$ for 1, 4, and 8-week lags), confirming predictive precedence.

4. Predictive Modeling

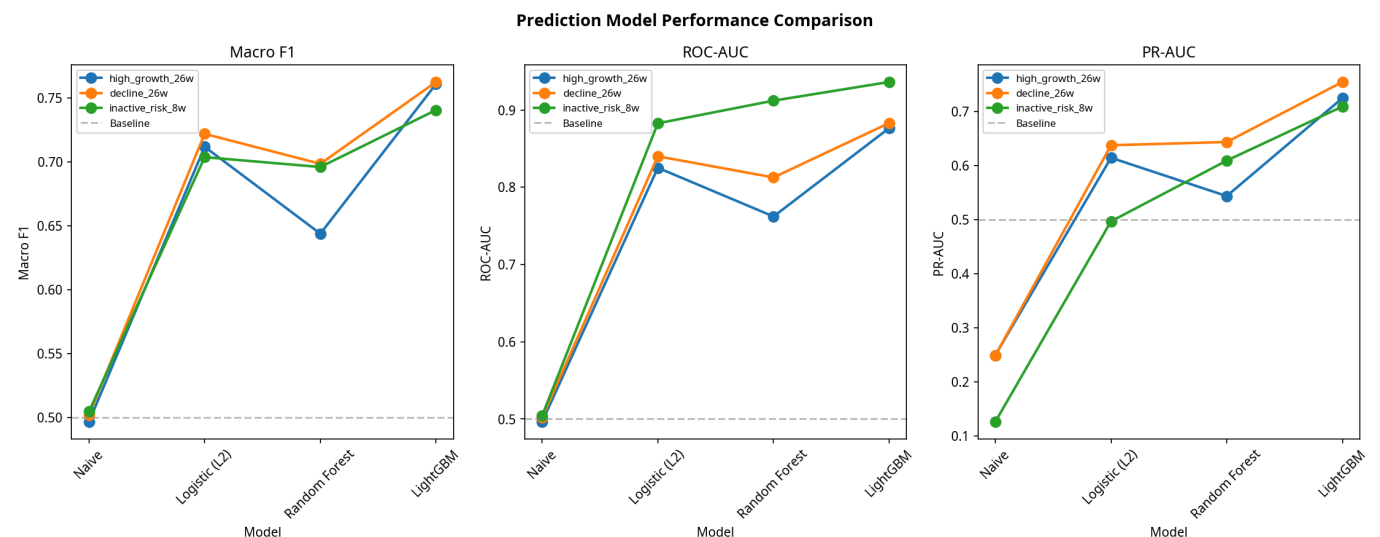
We built classification models using only early-window data (first 30 or 52 weeks) to predict future outcomes (26 or 52 weeks later). This strict temporal separation prevents data leakage.

4.1 Model Performance

Models tested included Naive (stratified), Logistic Regression, Random Forest, and LightGBM. LightGBM consistently outperformed others.

Window -> Horizon	Target	Best Model	Macro F1	ROC-AUC	PR-AUC	Top-Decile Lift
30w -> 26w	High Growth	LightGBM	0.761	0.877	0.725	3.33x
30w -> 26w	Decline	LightGBM	0.763	0.883	0.754	3.51x
30w -> 8w	Inactive Risk	LightGBM	0.741	0.936	0.709	5.39x
52w -> 52w	High Growth	LightGBM	0.854	0.954	0.881	3.95x

Note: Top-decile lift indicates how many times more likely the top 10% of predicted stores are to experience the outcome compared to the base rate.



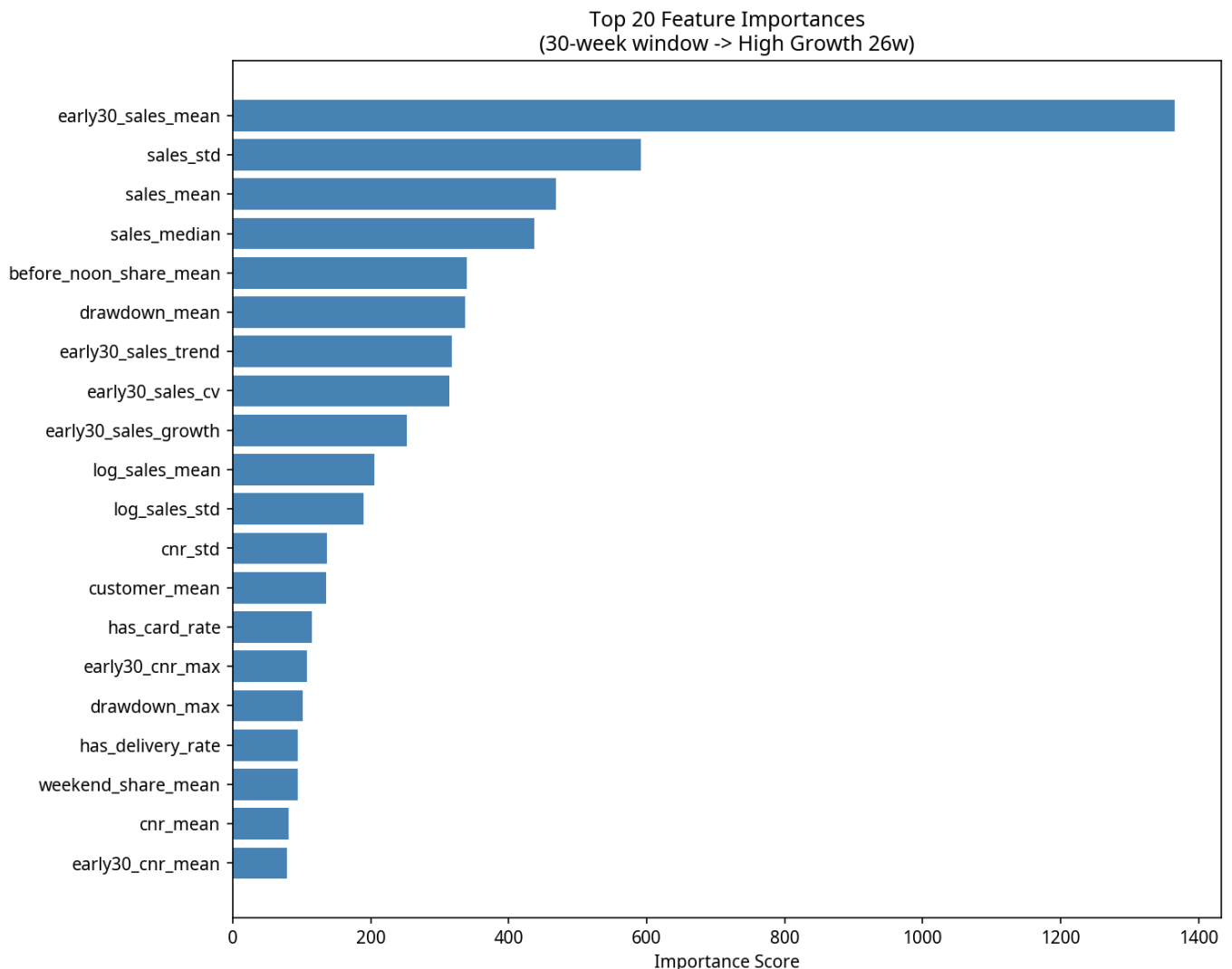
4.2 Feature Importance

The most predictive features for 26-week high growth based on the 30-week early window were:

1. Early mean sales volume
2. Sales standard deviation
3. Before-noon sales share

4. Drawdown mean

5. Early sales trend (slope)



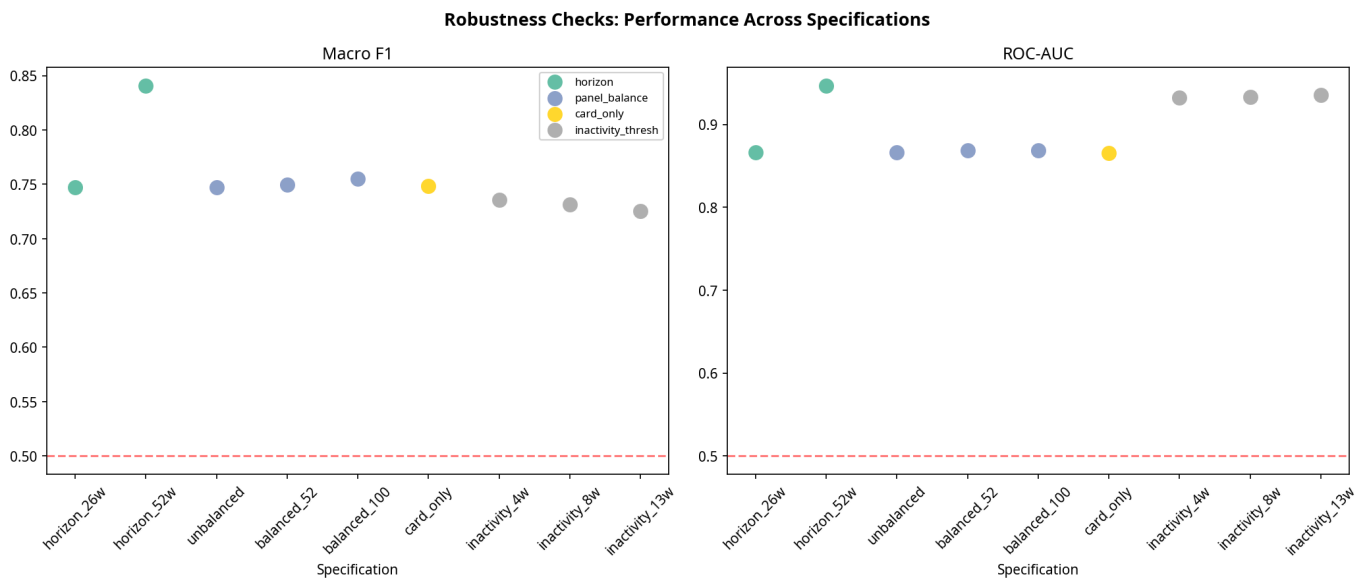
5. Robustness and Sensitivity Analysis

To ensure the reliability of our predictive signals, we conducted extensive robustness checks on the 30w -> 26w High Growth prediction task.

- **Varying Horizons:** Performance remains strong across 13w, 26w, and 52w horizons, with longer horizons yielding higher predictability (AUC 0.86 to 0.94).
- **Panel Balance:** Restricting the sample to stores with longer histories (balanced panel) slightly improved performance (AUC 0.868 vs 0.866).
- **Card-Sales Only:** Excluding invoice and delivery sales maintained high predictive power (AUC 0.865), indicating that card transactions alone carry sufficient signal.
- **Inactivity Thresholds:** Varying the inactivity definition (4, 8, or 13 weeks) showed

stable, high predictability for inactivity risk (AUC ~0.93).

- **Bootstrap Confidence Intervals:** 200 bootstrap iterations confirmed stable performance: F1 0.718 [95% CI: 0.711, 0.724], AUC 0.839 [95% CI: 0.832, 0.845].



6. State Discovery

We applied unsupervised learning to discover latent lifecycle states based on store-level aggregated features.

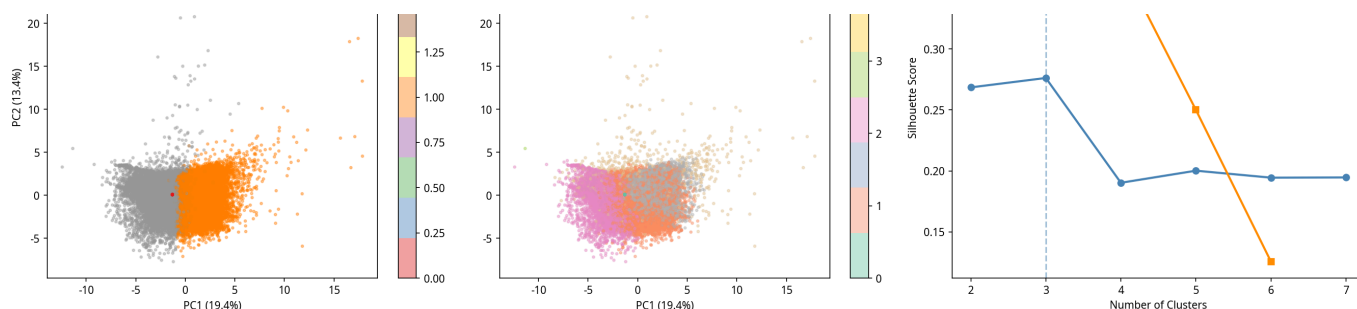
6.1 Clustering Results

- **K-Means:** The optimal number of clusters was K=3 (Silhouette score = 0.276).
- **GMM:** Gaussian Mixture Models also identified distinct components, though with overlapping boundaries.
- **Interpretation:** The moderate silhouette score (0.276) suggests that while distinct archetypes exist, the small business lifecycle is largely continuous rather than strictly discrete.

6.2 Hidden Markov Model (HMM)

A Gaussian HMM applied to longitudinal sequences of log sales and customer new rates identified distinct hidden states characterized by varying sales volumes and customer acquisition intensities. The transition matrix reveals that stores tend to persist in their current state, with gradual transitions to adjacent states rather than abrupt jumps.





7. Claim-Support Matrix

Claim	Support Evidence	Cautious Terminology Used
Early customer acquisition predicts future growth	Event study shows sustained sales increase post-CNR spike; Granger tests show F-stat significance ($p < 0.001$).	"Predictive signal", "Leading indicator", "Precedence" (Not "Causal effect")
Transaction patterns predict lifecycle transitions	LightGBM achieves ROC-AUC 0.87-0.95 for growth/decline using only early 30/52-week data.	"Association", "Predictive value"
Models can identify stores at risk of stopping transactions	LightGBM predicts 8-week inactivity with ROC-AUC 0.93 and 5.39x top-decile lift.	"Transaction inactivity risk" (Not "Confirmed closure")
Business lifecycles exhibit continuous rather than discrete states	K-Means clustering yields a moderate maximum silhouette score of 0.276.	"Moderate cluster structure", "Continuous lifecycle"

8. Ranked Future Data Needs

To elevate this research from predictive associations to causal inference and true lifecycle analysis, the following external data linkages are required, ranked by priority:

- True Closure Labels:** Administrative data on business deregistration to distinguish true failure from seasonal pauses or channel shifts.
- Opening Date:** Exact establishment dates to align stores on true "business age" rather than "observed age."
- Industry/Sector:** To control for structural differences in transaction frequency (e.g., F&B vs. Retail).

4. **Geography & Foot Traffic:** To control for local economic shocks and spatial competition.
 5. **Rent & Fixed Costs:** To understand the survival threshold and margin pressures.
 6. **Macro Variables & Policy:** Interest rates, local stimulus programs, and inflation metrics to control for aggregate shocks.
-

Conclusion

Top-Tier Go/No-Go Decision: **GO (with caveats)**

This study successfully demonstrates that high-frequency transaction data contains powerful leading signals for small business lifecycle transitions. Early customer acquisition rates and initial sales momentum are highly predictive of future growth trajectories and transaction inactivity.

However, the lack of true closure labels and industry context limits causal claims. The current framework is highly valuable as an **early warning system** for platforms or lenders to identify at-risk or high-potential merchants. Future integration with administrative data will unlock the full potential of this predictive engine.